Chapter 1

Introduction to Relational Databases in SQL

1.1 Uniquely identify records with key constraints

1.1.1 Keys and superkeys

What is a key?

- Attributes(s) that identify a record Uniquely.
- As long as attributes can be removed: **superkeys**.
- If no more attributes can be removed: minimal superkey or **key**.

license_no	**serial_no**	**make**	**model**	**year**
Texas ABC-739	A69352	Ford	Mustang	2
Florida TVP-347	B43696	Oldsmobile	Cutlass	5
New York MP0-22	X83554	Oldsmobile	Delta	1
California 432-TFY	C43742	Mercedes	190-D	99
California RSK-629	Y82935	Toyota	Camry	4
Texas RSK-629	U028365	Jaguar	XJS	4

Note: Table adapted from Elmasri, Navathe (2011): Fundamentals of Database Systems, 6th Ed., Pearson

SK1 = {license_no, serial_no, make, model, year}
SK2 = {license_no, serial_no, make, model}
SK3 = {make, model, year}, SK4 = {license_no, serial_no}, SKi, ..., SKn

A lot of possible superkeys with different combinations. But only four possible **keys**

K1 = {license_no}; K2 = {serial_no}; K3 = {model}; K4 = {make, year}

Here K1 to 3 only consist of one attribute. Removing either "make" or "year" from K4 would result in duplicates. Only one candidate key can be the chosen key.

A simple way of finding out whether a certain column(or combination) contains only unique values, wrap everything within the COUNT() function.

```
SELECT COUNT(column_a, column_b, ...)
FROM table_name;

-- then compare the number of rows, if the number of Distinct row matches the total number of
```

```sql
--rows then the column or combination of column can be used as keys

SELECT COUNT(DISTINCT(column_a, column_b, ...))
```

1.2 Primary keys

- One primary key per database table, chosen from candidate keys
- Uniquely identifies records, e.g. for referencing in other tables
- Unique and not-null constraint both apply
- Primary keys are time-invariant: choose columns wisely

1.2.1 Specifying primary keys

```sql
CREATE TABLE products (
    product_no integer UNIQUE NOT NULL, -- same constraint as of Primary Key
    name text,
    price numberic
);

CREATE TABLE products (
    product_no integer PRIMARY KEY,
    name text,
    price numberic
);

-- Designmate composite key as primary key
CREATE TABLE example (
    a integer,
    b integer,
    c integer,
    PRIMARY KEY (a, c)
);
```

Note: Taken from PostgreSQL documentation.

1.2.2 Adding primary key to existing table

```sql
ALTER TABLE table_name
ADD CONSTRAINT some_name PRIMARY KEY (column_name)
```

1.3 Surrogate keys